JetFUEL: Jet Flow Matching with Equivariance Under the Lorentz Group

Abstract

We introduce JetFUEL (Jet Flow Matching with Equivariance Under the Lorentz Group), a Lorentz-equivariant message-passing vector field trained by Riemannian flow matching on a regulated mass shell. The network uses evolving auxiliary Lorentz vectors for message passing, while its output field is built intrinsically from mass-shell logarithmic maps and normalized tangent projections of two condition-derived reference vectors. This document specifies the geometry, network, conditional prior, coupling, objective, and sampling procedure.

1 Geometry and notation

Four-vectors are ordered as $p = (p^0, p^1, p^2, p^3) = (E, p_x, p_y, p_z) \in \mathbb{R}^{1,3}$ and use metric signature $(+, -, -, -)$,

$$\langle p, q\rangle_M = p^0 q^0 - \sum_{a=1}^{3} p^a q^a, \qquad \|p\|_M^2 \equiv p^2 = \langle p, p\rangle_M. \tag{1}$$

For a fixed radius (regulator mass) $m > 0$, the forward hyperboloid is

$$\mathcal{H}_m = \{p \in \mathbb{R}^{1,3} : p^2 = m^2,\ p^0 > 0\}. \tag{2}$$

This is the radius-m version of the unit hyperboloid used by Lou et al. [1, Table 2]; their convention is obtained with $x = p/m$ and $\langle x, y\rangle_L = -\langle p, q\rangle_M/m^2$.

The tangent space and induced Riemannian metric of this hyperboloid are the standard expressions [1, Table 2]:

$$T_p\mathcal{H}_m = \{u : \langle p, u\rangle_M = 0\}, \qquad g_p(u, v) = -\langle u, v\rangle_M, \qquad \|u\|_p = \sqrt{-\langle u, u\rangle_M}. \tag{3}$$

The corresponding ambient-to-tangent projection is [1, Table 2]

$$\Pi_p(v) = v - \frac{\langle p, v\rangle_M}{m^2} p. \tag{4}$$

For $p, q \in \mathcal{H}_m$, the hyperbolic distance is [1, Table 2]

$$\gamma(p, q) = \frac{\langle p, q\rangle_M}{m^2}, \qquad d_m(p, q) = m \operatorname{arcosh}\big(\gamma(p, q)\big). \tag{5}$$

The associated logarithmic and exponential maps are [1, Table 2]

$$\operatorname{Log}_p(q) = \frac{\operatorname{arcosh}(\gamma(p, q))}{\sinh(\operatorname{arcosh}(\gamma(p, q)))}\big(q - \gamma(p, q)p\big), \tag{6}$$

$$\operatorname{Exp}_p(u) = \cosh\left(\frac{\|u\|_p}{m}\right) p + m \sinh\left(\frac{\|u\|_p}{m}\right) \frac{u}{\|u\|_p}. \tag{7}$$

Both expressions use their continuous values at $q = p$ and $u = 0$.

Geometric interpretation. With $w_{p\to q} = q - \gamma(p, q)p$ and $\|w_{p\to q}\|_p = m \sinh(\operatorname{arcosh} \gamma(p, q))$, the same map is $\operatorname{Log}_p(q) = d_m(p, q)w_{p\to q}/\|w_{p\to q}\|_p$. Thus the log map is the unit tangent direction from p toward q, scaled by the geodesic distance.

A jet with N real constituents is modelled as a point on the standard product manifold. Its product metric and componentwise maps are likewise standard [2]:

$$\mathcal{M}_N = \mathcal{H}_m^N, \qquad G_X(U,V) = \sum_{i=1}^{N} g_{x_i}(u_i, v_i), \tag{8}$$

where $X = (x_1, \ldots, x_N)$ and $U = (u_1, \ldots, u_N) \in T_X\mathcal{M}_N$. The product maps act componentwise:

$$\mathrm{Exp}_X(U) = (\mathrm{Exp}_{x_1}(u_1), \ldots, \mathrm{Exp}_{x_N}(u_N)), \qquad \mathrm{Log}_X(Y) = (\mathrm{Log}_{x_1}(y_1), \ldots, \mathrm{Log}_{x_N}(y_N)). \tag{9}$$

The choice to use this product manifold for jet constituents, and the masking of padded slots in implementation, are model-specific conventions.

2 Conditional inputs

The vector field receives the current physical state $X \in \mathcal{M}_N$, flow time $t \in [0,1]$, and condition $c = [c_{\text{type}}, N, p_{T,J}, m_J]$. It also receives the ordered references $r_0 = e_t = (1,0,0,0)$ and $r_1 = p_J$. The latter is constructed from the conditioned jet attributes,

$$p_J = \frac{1}{s}\begin{pmatrix} m_{T,J}\cosh\eta_J \\ p_{T,J}\cos\phi_J \\ p_{T,J}\sin\phi_J \\ m_{T,J}\sinh\eta_J \end{pmatrix}, \qquad m_{T,J} = \sqrt{p_{T,J}^2 + m_J^2}, \tag{10}$$

where s is the dataset momentum scale. These vectors are condition-derived; they are not formed by summing the target constituents.

3 JetFUEL vector field

The network maintains invariant scalar features $h_i^\ell \in \mathbb{R}^{d_h}$ and auxiliary Lorentz-vector features $y_i^\ell \in \mathbb{R}^{1,3}$, with $d_h = 96$ and $L = 6$ layers. The physical state x_i remains fixed during a network evaluation. Only y_i^ℓ evolves through message passing. The graph is complete over real constituents, with $\mathcal{N}(i) = \{j : j \neq i\}$.

Algorithm 1 JetFUEL vector field $V_\theta(X, t, c; r_0, r_1)$

Require: $X = (x_1, \dots, x_N)$, t, c, (r_0, r_1)
Ensure: $V_\theta(X, t, c; r_0, r_1) \in T_X\mathcal{M}_N$

$y_i^0 \leftarrow x_i$ and $\tau(t) \leftarrow [t, \sin(\pi t), \cos(\pi t)]$ for all i
$h_i^0 \leftarrow \phi_{\text{seed}}([\psi(\langle y_i^0, r_0\rangle_M), \psi(\langle y_i^0, r_1\rangle_M)]) + \phi_c(c) - \phi_t(\tau(t))$
for $\ell = 0, \dots, L-1$ **do**
  $\bar{h}_i^\ell \leftarrow \text{LayerNorm}(h_i^\ell)$
  for each $j \in \mathcal{N}(i)$ **do**
    $\xi_{ij}^\ell \leftarrow [\psi(\|y_i^\ell - y_j^\ell\|_M^2), \psi(\langle y_i^\ell, y_j^\ell\rangle_M)]$
    $M_{ij}^\ell \leftarrow \phi_e^\ell(\bar{h}_i^\ell, \bar{h}_j^\ell, \xi_{ij}^\ell)$; $\alpha_{ij}^\ell \leftarrow \phi_x^\ell(M_{ij}^\ell)$
  end for
  $h_i^{\ell+1} \leftarrow h_i^\ell + \phi_h^\ell\left(\bar{h}_i^\ell, |\mathcal{N}(i)|^{-1/2} \sum_{j\in\mathcal{N}(i)} M_{ij}^\ell\right)$
  $y_i^{\ell+1} \leftarrow y_i^\ell + \varepsilon_y |\mathcal{N}(i)|^{-1/2} \sum_{j\in\mathcal{N}(i)} \alpha_{ij}^\ell (y_i^\ell - y_j^\ell)$
end for
for each $j \in \mathcal{N}(i)$ **do**
  $\xi_{ij}^L \leftarrow [\psi(\|y_i^L - y_j^L\|_M^2), \psi(\langle y_i^L, y_j^L\rangle_M)]$
  $a_{ij} \leftarrow \phi_f(h_i^L, h_j^L, \xi_{ij}^L)$; $q_{ij} \leftarrow \text{Log}_{x_i}(x_j)/(m + d_m(x_i, x_j))$
end for
$v_i^{\text{part}} \leftarrow |\mathcal{N}(i)|^{-1/2} \sum_{j\in\mathcal{N}(i)} a_{ij} q_{ij}$
$(b_{i0}, b_{i1}) \leftarrow \phi_r(h_i^L, [\psi(\langle y_i^L, r_0\rangle_M), \psi(\langle y_i^L, r_1\rangle_M)])$
$q_{ir}^{\text{ref}} \leftarrow \Pi_{x_i}(r_r)/(m + \|\Pi_{x_i}(r_r)\|_{x_i})$, for $r \in \{0, 1\}$
$v_i^{\text{ref}} \leftarrow b_{i0} q_{i0}^{\text{ref}} + b_{i1} q_{i1}^{\text{ref}}$
$v_{\theta,i} \leftarrow v_i^{\text{part}} + v_i^{\text{ref}} \in T_{x_i}\mathcal{H}_m$
return $V_\theta = (v_{\theta,1}, \dots, v_{\theta,N})$

The signed logarithmic preprocessing in Algorithm 1 is

$$\psi(z) = \text{sgn}(z)\log(1 + |z|). \tag{11}$$

All learned maps $\phi_\bullet$ act on scalar invariant inputs. With $\sigma = \text{SiLU}$, their widths are

$$\phi_c : d_c \to d_h \to d_h, \qquad \phi_t : 3 \to d_h \to d_h, \qquad \phi_{\text{seed}} : 2 \to d_h \to d_h, \tag{12}$$

$$\phi_e^\ell : (2d_h + 2) \to d_h \to d_h, \qquad \phi_h^\ell : 2d_h \to 2d_h \to d_h, \tag{13}$$

$$\phi_x^\ell : d_h \to d_h \to 1, \qquad \phi_f : (2d_h + 2) \to d_h \to 1, \qquad \phi_r : (d_h + 2) \to d_h \to 2. \tag{14}$$

Every hidden linear layer is followed by σ; ϕ_e^ℓ also applies σ after its second linear layer. The final layers of ϕ_x^ℓ, ϕ_f, and ϕ_r have weights initialized from $\mathcal{N}(0, 10^{-6})$ and zero bias. The auxiliary-vector residual scale is $\varepsilon_y = 10^{-2}$.

There is no message gate, softmax, absolute-value operation, or positivity constraint. Thus the graph aggregation and the final particle readout are signed sums. The particle term is intrinsically tangent because $q_{ij} \in T_{x_i}\mathcal{H}_m$, and the reference term is tangent by Eq. (4). Therefore their sum is the model output directly in $T_{x_i}\mathcal{H}_m$.

For $\Lambda \in SO^+(1, 3)$, the network is jointly Lorentz equivariant,

$$V_\theta(\Lambda^{\times N} X, t, c; \Lambda r_0, \Lambda r_1) = \Lambda^{\times N} V_\theta(X, t, c; r_0, r_1). \tag{15}$$

The statement follows because MLP inputs are Lorentz scalars, learned vector coefficients are scalars, and tangent maps commute with Lorentz transformations. Holding the references fixed in the laboratory frame supplies conditional symmetry breaking through the beam/time and jet directions.

4 Mass-shell Riemannian flow matching

4.1 Source and data distributions

Observed massless constituent momenta $\bar{x}_i = (\|\boldsymbol{p}_i\|_2, \boldsymbol{p}_i)$ are lifted to the shell by preserving spatial momentum,

$$\iota_m(\bar{x}_i) = \left(\sqrt{\|\boldsymbol{p}_i\|_2^2 + m^2}, \boldsymbol{p}_i\right). \tag{16}$$

Let $X_1 \sim q_1(\cdot \mid c)$ denote a lifted target jet. The conditional source distribution $q_0(\cdot \mid c)$ is an axis-aligned lognormal prior. It samples

$$\Delta\eta_i, \Delta\phi_i \overset{\text{iid}}{\sim} \mathcal{N}(0, 0.15^2), \qquad z_i \overset{\text{iid}}{\sim} \mathcal{N}(0,1), \tag{17}$$

$$w_i = \frac{e^{z_i}}{\sum_j e^{z_j}}, \qquad \eta_i = \eta_J + \Delta\eta_i, \qquad \phi_i = \phi_J + \Delta\phi_i. \tag{18}$$

Writing

$$R = \left\| \sum_i w_i (\cos\phi_i, \sin\phi_i) \right\|_2, \qquad p_{T,i} = w_i \frac{p_{T,J}}{R}, \tag{19}$$

the unlifted source four-vector is

$$\bar{x}_{0,i} = \frac{1}{s}\left(p_{T,i}\cosh\eta_i,\ p_{T,i}\cos\phi_i,\ p_{T,i}\sin\phi_i,\ p_{T,i}\sinh\eta_i\right), \tag{20}$$

followed by the lift in Eq. (16).

4.2 Online geodesic coupling

Particle indices do not identify physical objects. For every fresh source draw, prior constituents are paired with target constituents through the per-jet Hungarian assignment

$$\pi^\star = \arg\min_{\pi \in S_N} \sum_{i=1}^{N} d_m(x_{0,\pi(i)}, x_{1,i})^2. \tag{21}$$

The source is permuted according to $\pi^\star$ before the RFM path is formed. This coupling is recomputed online for each training batch and does not reuse a fixed noise-to-data pairing.

4.3 Conditional geodesic path and objective

For a coupled pair (X_0, X_1) and $t \sim \mathcal{U}[0,1]$, RFM uses the product-manifold geodesic

$$X_t = \mathrm{Exp}_{X_1}\left((1-t)\,\mathrm{Log}_{X_1}(X_0)\right) = \mathrm{Exp}_{X_0}\left(t\,\mathrm{Log}_{X_0}(X_1)\right). \tag{22}$$

Its conditional velocity, tangent at X_t, is

$$U_t(X_t \mid X_1) = \frac{\mathrm{Log}_{X_t}(X_1)}{1-t} \in T_{X_t}\mathcal{M}_N. \tag{23}$$

Equation (23) is the geodesic specialization of the distance-gradient target in Chen and Lipman [2, Eqs. (13)–(15)], not a separate choice of target field. Let $d_\mathcal{M}$ denote the geodesic distance of the product metric. Along the conditional geodesic,

$$d_\mathcal{M}(X_t, X_1) = (1-t)\,d_\mathcal{M}(X_0, X_1), \qquad \|\nabla_{X_t} d_\mathcal{M}(X_t, X_1)\|_{X_t} = 1, \tag{24}$$

and the standard distance-gradient identity is

$$\mathrm{Log}_{X_t}(X_1) = -d_\mathcal{M}(X_t, X_1)\nabla_{X_t} d_\mathcal{M}(X_t, X_1). \tag{25}$$

For $\kappa(t) = 1-t$, the Chen–Lipman target therefore reduces as

$$\begin{aligned} U_t(X_t \mid X_1) &= -d_\mathcal{M}(X_0, X_1)\nabla_{X_t} d_\mathcal{M}(X_t, X_1) \\ &= \frac{\mathrm{Log}_{X_t}(X_1)}{1-t}. \end{aligned} \tag{26}$$

The numerator vanishes linearly as $t \to 1$; implementation terminates just before $t = 1$ and uses a protected denominator.

The model is trained by the product-manifold squared error

$$\mathcal{L}_{\mathrm{RFM}}(\theta) = \mathbb{E}\left[\|V_\theta(X_t, t, c; r_0, r_1) - U_t(X_t \mid X_1)\|^2_{X_t}\right], \tag{27}$$

where the norm is defined by Eq. (8) and is averaged over real particles in implementation. Under squared-error regression, the population optimum is the marginal velocity

$$V^\star(x, t, c) = \mathbb{E}[U_t(X_t \mid X_1) \mid X_t = x, c]. \tag{28}$$

This field has the same time marginals as the mixture of conditional geodesic paths, so integrating it transports $q_0(\cdot \mid c)$ toward $q_1(\cdot \mid c)$.

4.4 Sampling

Generation starts from $X(0) \sim q_0(\cdot \mid c)$ and integrates

$$\dot{X}(t) = V_\theta(X(t), t, c; r_0, r_1). \tag{29}$$

Using geodesic Euler integration, one step is

$$X_{k+1} = \mathrm{Exp}_{X_k}(\Delta t\, V_\theta(X_k, t_k, c; r_0, r_1)). \tag{30}$$

The product exponential is applied constituentwise. Hence all generated particles remain on the positive-energy mass shell throughout sampling. The reported Arm H configuration uses $m = 0.1$, 64 Euler steps, and an endpoint time of 0.99999.

5 Ablation results

Unless stated otherwise, the endpoint metrics below use EMA weights, 50,000 generated jets, and 64 geodesic Euler steps. FPND is computed after the validated JetNet coordinate conversion; historical FPND values produced before that repair are excluded. W1M, W1P, and FPD are reported in the native normalized JetNet convention. These are single-run estimates, so small differences should not be interpreted as seed-level significance.

5.1 Geometry and readout construction

The earliest controlled comparison replaced ambient Euclidean interpolation with the regulated mass-shell path while holding the presence or absence of reference readout fixed. Table 1 reports only quantities that remain valid for those historical runs. The mass-shell formulation reduced W1M by approximately an order of magnitude and removed the large spacelike component population. Physical support is therefore enforced by construction rather than learned as an approximate constraint.

Table 1: Early geometry ablation on gluon-30 at 200k updates. FPND is omitted because these runs predate the corrected ParticleNet input convention. Percentages refer to generated jets or generated constituents as appropriate.

Arm	Geometry and readout	W1M ↓	Invalid (%)	Spacelike (%)	$E < 0$ (%)
A	Euclidean, no references	0.249857	4.492	25.45	2.290
B	Mass shell, no references	0.025672	0.000	0.00	0.000
C	Euclidean, plain references	0.105699	0.006	33.87	0.298
D	Mass shell, plain references	0.008915	0.004	0.00	0.000

The later G–J campaign isolates normalized tangent references versus raw reference projections and evolving auxiliary geometry versus fixed physical geometry. Table 2 shows a consistent advantage for evolving y^ℓ over fixed geometry (G versus I and H versus J). Reference normalization is mildly favorable in the evolving model but not decisive at this sample size. Reinjecting condition and time embeddings in every block does not help.

Table 2: Controlled gluon-30 construction ablations. "Evol." denotes evolving auxiliary Lorentz vectors and "fixed" denotes message geometry frozen to x. The 500k rows test training duration rather than a new architecture.

Run	Construction	$\mathrm{FPND}_g \downarrow$	FPD $\times 10^4 \downarrow$	W1M $\times 10^3 \downarrow$	mean W1P $\times 10^3 \downarrow$	Cov. $\uparrow$	MMD $\times 10^2 \downarrow$
G, 200k	raw references, evol.	0.3935	1.593	1.884	1.371	0.563	2.647
H, 200k	normalized references, evol.	**0.3745**	**1.469**	**1.808**	**1.264**	**0.565**	2.684
I, 200k	raw references, fixed	0.4264	1.685	1.981	1.405	0.545	2.793
J, 200k	normalized references, fixed	0.4563	1.747	2.191	1.465	0.539	2.815
H, 200k	per-block condition/time	0.3868	1.524	2.164	1.299	0.544	2.740
G, 500k	raw references, evol.	0.3754	**1.389**	**1.868**	1.384	**0.555**	**2.676**
H, 500k	normalized references, evol.	**0.3678**	1.397	2.030	**1.334**	0.551	2.844

5.2 Coupling and the auxiliary workspace

Table 3 is the matched 994k-update gluon-30 training comparison. Removing online geodesic ICP increases FPND_g from 0.391 to 1.054 and worsens FPD, W1M, and W1P while retaining exact physical support. ICP therefore improves the learned distribution rather than merely preventing invalid trajectories. Replacing the physical log-map directions by raw latent displacements $y_j^L - y_i^L$ leaves the already-degraded no-ICP result nearly unchanged. Because the completed latent run also removes ICP, it does *not* isolate the readout choice under the canonical coupling; the corresponding ICP-enabled retraining remains necessary before simplifying the final physical readout.

Table 3: Matched 994k-update gluon-30 coupling/readout runs. "Physical" uses coefficients from (h^L, y^L) but directions from $\mathrm{Log}_{x_i}(x_j)$; "latent" uses $y_j^L - y_i^L$ for the directions.

Coupling	Directions	$\mathrm{FPND}_g \downarrow$	FPD $\times 10^4 \downarrow$	W1M $\times 10^3 \downarrow$	mean W1P $\times 10^3 \downarrow$	Cov. $\uparrow$	MMD $\times 10^2 \downarrow$
online ICP	physical	**0.3910**	**1.448**	**2.025**	**1.529**	**0.556**	2.690
none	physical	1.0541	4.859	3.932	2.763	0.536	**2.630**
none	latent	1.0459	4.837	3.918	2.825	0.553	2.758

5.3 Rejected auxiliary mechanisms and capacity reduction

The matched 100k-update Gram-auxiliary experiment in Table 4 improves a few one-dimensional quantities but damages the joint 36-EFP FPD by a factor of 4.3 and worsens W1P. It was therefore rejected. The 2k token smoke tests in Table 5 likewise show no loss advantage; the scalar-vector token violates the endpoint-invalidity gate. Neither mechanism is present in the final model.

Table 4: Matched gluon-30 auxiliary-objective ablation at 100k updates. Historical FPND is omitted because these runs predate the corrected input convention.

Objective	W1M $\times 10^3$	mean W1P $\times 10^3$	mean W1EFP $\times 10^5$	FPD $\times 10^4$	Cov.	MMD $\times 10^2$
control	1.805	**0.959**	1.487	**6.15**	0.54	2.681
Gram-50	**1.724**	1.137	**1.323**	26.40	**0.56**	**2.619**

Table 5: Jet-token qualification smoke test at 2k optimizer updates.

Token	Final median loss ↓	Invalid endpoints	Qualification
none	**0.004798**	0/2000	pass
scalar	0.005057	1/2000	pass with warning
scalar+vector	—	3/2000	fail

The subsequent backbone simplification reduced parameter count by more than an order of magnitude without a corresponding collapse in corrected FPND. Table 6 is a capacity frontier rather than a perfectly matched width ablation: variants a–c use width 96, d uses width 128, and the historical attention model uses width 256. The readout-only, no-pooling variant is the smallest and best of the message-passing variants on FPND; global pooling is not justified by these results.

Table 6: Gluon-30 capacity and message-passing simplification at 600k updates. The attention-model FPND was recomputed after the coordinate repair; its other historical metrics are omitted here.

Backbone	Parameters	FPND_g ↓	FPD $\times 10^4$ ↓	W1M $\times 10^3$ ↓	Cov. ↑	MMD $\times 10^2$ ↓
tangent attention, width 256	7.39M	**0.507**	—	—	—	—
readout-only, no pool, width 96	0.513M	0.5295	3.484	2.431	0.550	**2.591**
tangent channels, no pool, width 96	0.560M	0.5504	2.995	**2.310**	0.536	2.744
readout-only, global pool, width 96	0.848M	0.5460	2.760	2.602	**0.552**	2.704
tangent channels, global pool, width 128	0.910M	0.553	**2.462**	2.635	0.548	2.704

5.4 Training stability and readout initialization

Zero-initializing the terminal velocity heads makes the initial sampler the identity map. The 2k-update qualification in Table 7 shows that initialization strongly changes the route to a stable solution, but the terminal losses and endpoint validity are tied. Zero initialization was retained as the safer default; this is not a claim of better converged sample quality. The experiment also demonstrates why loss alone is insufficient for checkpoint selection: either arm can have an ordinary loss while fixed-seed sampler probes remain unstable.

Table 7: Velocity-readout initialization qualification. Probe entries report unstable trajectories out of 64.

Quantity	small-normal	zero
unstable at step 0	64/64	**0/64**
unstable at step 100	64/64	**0/64**
unstable at step 500	**60/64**	63/64
unstable at step 1000	**0/64**	18/64
unstable at step 2000	0/64	0/64
first-200 median loss	7.664803	**0.015781**
final-200 median loss	0.004732	**0.004675**
final failures (of 128)	0	0

5.5 Inference-time simplifications and scaling

The GQT-994k checkpoint gives a clean integration-resolution curve without retraining. As shown in Table 8, doubling from 32 to 64 and from 64 to 128 steps improves FPND in every class while approximately doubling generation time. The continued gain at 128 steps indicates discretization error at the canonical 64-step setting rather than a saturated sampler.

Table 8: Geodesic Euler scaling for the GQT-994k checkpoint at fixed $w = 0$. Generation time is for 150,000 class-stratified samples on the same A40 inference setup.

Steps	$\text{FPND}_g \downarrow$	$\text{FPND}_q \downarrow$	$\text{FPND}_t \downarrow$	Generation time (s)
32	0.4059	0.6132	0.9646	992.1
64	0.1420	0.2679	0.5244	1853.8
128	**0.0797**	**0.1546**	**0.3693**	3713.5

The class-conditional guidance sweep in Table 9 is sharply class-dependent. Guidance is unnecessary for gluons, while light-quark and top jets prefer $w = 0.25$ and $w = 0.5$, respectively. Larger weights rapidly degrade all classes.

Table 9: Classwise FPND for the GQT-994k classifier-free-guidance sweep at 64 Euler steps. Bold entries are the selected per-class deployment settings.

w	$\text{FPND}_g \downarrow$	$\text{FPND}_q \downarrow$	$\text{FPND}_t \downarrow$
0.00	**0.1420**	0.2679	0.5244
0.25	0.3182	**0.2137**	0.2801
0.50	0.5255	0.3138	**0.2019**
0.75	0.7063	0.5796	0.2582
1.00	0.9160	1.6340	0.4261
2.00	4.5388	5.7217	2.1008
3.00	25.0270	15.1774	8.4812

Finally, Table 10 tests the redundant terminal tangent projection that previously followed the sum of already-tangent particle and reference terms. Turning it off leaves FPND unchanged to approximately six significant figures, and the 36-EFP FPD is numerically identical. The projection was therefore deleted from both implementation and algorithm.

Table 10: Final-projection ablation on the GQT-994k checkpoint at the selected per-class guidance weights. "On" applies Π_{x_i} to the final sum; "off" returns that already-tangent sum directly.

Class (w)	FPND on	FPND off	FPD on $\times 10^4$	FPD off $\times 10^4$
g (0)	0.14199731	0.14199728	8.7697	8.7697
q (0.25)	0.21368916	0.21369031	33.2823	33.2823
t (0.5)	0.20192473	0.20192409	24.2870	24.2870

References

[1] Aaron Lou, Derek Lim, Isay Katsman, Leo Huang, Qingxuan Jiang, Ser-Nam Lim, and Christopher M. De Sa. Neural Manifold Ordinary Differential Equations. In *Advances in Neural Information Processing Systems*, 2020. Appendix B.4, Table 2.

[2] Ricky T. Q. Chen and Yaron Lipman. Flow Matching on General Geometries. *arXiv preprint arXiv:2302.03660*, 2023.